

Violet Yinuo Han

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RESEARCH INTEREST

Keywords: Physical AI, Human-Robot Interaction, User Modeling, Intelligent Environments

My research builds physical AI systems that understand and adapt to users in everyday environments.

I develop (1) robot learning frameworks that discover user preferences from behavioral observation, enabling personalized interaction without constant explicit instruction, and (2) robotic interfaces that integrate into existing objects, enabling manipulation, movement, and tactile feedback.

EDUCATION

PhD Student, Carnegie Mellon University, School of Computer Science advised by Professor Alexandra Ion	Started August 2023
M.S., Carnegie Mellon University <i>Thesis:</i> Building a Bidirectional Bridge Between the Digital and Physical Worlds	Graduated May 2023
B.F.A., University of Michigan <i>Thesis:</i> 86 Billions: Experiencing the Human Brain From a Neuron's Perspective	Graduated May 2020

PUBLICATIONS

- [1] **Towards Unobtrusive Physical AI: Augmenting Everyday Objects with Intelligence and Robotic Movement for Proactive Assistance.**
Violet Yinuo Han, Jesse T. Gonzalez, Christina Yang, Zhiruo Wang, Scott E. Hudson, Alexandra Ion. In Proceedings of UIST'25, Busan, Korea. Sept. 28 - Oct. 1, 2025.
(22.2% acceptance rate)
- [2] **Transforming Everyday Objects into Dynamic Interfaces using Smart Flat-Foldable Structures.**
Violet Yinuo Han, Amber Yinglei Chen, Mason Zadan, Jesse T. Gonzalez, Dinesh K. Patel, Wendy Fangwen Yu, Carmel Majidi, Alexandra Ion. In Proceedings of UIST'25, Busan, Korea. Sept. 28 - Oct. 1, 2025.
(22.2% acceptance rate)
- [3] **A Dynamic Bayesian Network Based Framework for Multimodal Context-Aware Interactions.**
Violet Yinuo Han, Tianyi Wang, Hyunsung Cho, Kashyap Todi, Ajoy Savio Fernandes, Andre Levi, Zheng Zhang, Tovi Grossman, Alexandra Ion, Tanya R Jonker. In Proceedings of IUI'25, Cagliari, Italy. March 24 - 27, 2025.
(25% acceptance rate) **Selected for Journal Extension**
- [4] **Robotic Metamaterials: A Modular System for Hands-On Configuration of Ad-Hoc Dynamic Applications.**
Zhitong Cui, Shuhong Wang, **Violet Yinuo Han**, Tucker Rae-Grant, Willa Yunqi Yang, Alan Zhu, Scott E Hudson, Alexandra Ion. In Proceedings of CHI'24, Honolulu, HI. May.11 - 16, 2024.
(26.4% acceptance rate)
- [5] **Parametric Haptics: Versatile Geometry-based Tactile Feedback Devices.**
Violet Yinuo Han, Abena Boadi-Agyemang, Yuyu Lin, David Lindlbauer, Alexandra Ion. In Proceedings of UIST'23, San Francisco, CA. Oct. 29 - Nov. 1, 2023.
(21% acceptance rate)
- [6] **BlendMR: A Computational Method to Create Ambient Mixed Reality Interfaces.**
Violet Yinuo Han, Hyunsung Cho, Kiyosu Maeda, Alexandra Ion, David Lindlbauer. In Proceedings of ISS'23, Pittsburgh, PA. Nov. 5 - Nov. 8, 2023.
(24-28% acceptance rate) **Best Paper Award -Top 1**
- [7] **Permeable Thermistor Temperature Sensors Based on Porous Melamine Foam.**
Hugo de Souza Oliveira, Niloofar Saeedzadeh Khaanghah, **Violet Yinuo Han**, Alejandro Carrasco-Pena, Alexandra Ion, Michael Haller, Giuseppe Cantarella, Niko Münzenrieder. In IEEE Sensors Letters, Vol. 7, No. 5, pp. 1-4, Art no. 2500904, May 2023.
(20-30% acceptance rate)

TEACHING EXPERIENCE

- Guest Lecturer at Carnegie Mellon University** April 2025
Computational Methods for Interactive Systems (05-499/899), Prof. Alexandra Ion
- Guest lecture on using dynamic Bayesian networks to model interactions for building interactive systems.
- Teaching Assistant at Carnegie Mellon University** August - December 2022
Computational Perception (15-387/86-375/675), Prof. Tai Sing Lee
- This course investigates human visual perception with computational models. I hosted weekly office hours, created, updated, and graded homework assignments. I created lecture notes to better facilitate students' learning.

PROFESSIONAL EXPERIENCE

- Research Scientist Intern, Meta Reality Labs** May - September 2024
- Proposed a dynamic Bayesian network based framework for multimodal context-aware interactions. Led to a publication at IUI'25.
- System Experience Design Intern, OnePlus** May - July 2021
- Worked with designers and engineers to maintain and update OnePlus OxygenOS.
- Interaction Design Intern, WhaleDynamic** Feb - May 2021
- Designed interactions for an autonomous delivery vehicle and presented a prototype at the Shanghai International Automobile Industry Exhibition.

AWARDS AND GRANTS

- Qualcomm Innovation Fellowship Finalist** (with Weiwei Sun, competing for winner selection) 2026
- Special Recognition for Outstanding Review** ACM UIST'25 2025
- Best Paper Award** (Top 1) ACM ISS'23 for *BlendMR* [7] 2023
- Snap Creative Challenge Award** (10k) with Prof. David Lindlbauer and Dr. Yukang Yan 2022
- Carnegie Mellon University SoA Commendation** 2022
- Carnegie Mellon University SoA Merit Scholarship** (10k) 2021
- University of Michigan University Honors** 2020
- University of Michigan University Honors** 2018

MEDIA COVERAGE

- ComputerWorld** "Why ambient robots beat humanoid robots " [4] 2026
- CNET** "Robot Coffee Cups? Self-Driving Trivets? AI Researchers Made It Happen" [4] 2025
- TechXplore** "A stapler that knows when you need it: Using AI to turn everyday objects into proactive..." [4] 2025
- The Brighter Side of News** "AI revolution: Researchers teach everyday objects to sense, think, and move" [4] 2025

MENTORING

- Dian Zhu**, Math at CMU, Undergraduate Research Assistant Fall 2025
- Narayan Ashanahalli**, Architecture at CMU, Masters Research Assistant Fall 2025
- Aiden Li**, CS at UCL, Undergraduate Research Assistant Summer 2025
- Invited talk at the Ellis High School**, Pittsburgh April 2025
- Christina Yang**, Design & HCI at CMU, Masters Research Assistant Fall 2024
- Anagha Srinivasan**, CS at CMU, Undergraduate Research Assistant Spring 2024
- Ziyu Li**, ECE at CMU, Undergraduate Research Assistant Fall 2023
- Wendy Yu**, Design & HCI at CMU, Undergraduate Research Assistant Fall 2023, Spring 2024

ACADEMIC SERVICES

- UIST Reviewer** 2024, 2025
- CHI Reviewer** 2024, 2025
- CHI LBW Reviewer** 2023, 2024
- CHI Assistant to Session Chair** 2024
- UIST Student Volunteer** 2022